First, whereas a cosmological constant is constant—it does not vary with time, so it provides a constant, unchanging outward push—a supercooled Higgs field need not be constant. Think of a frog perched on the bump in Figure 10.1a. It may hang out there for a while, but sooner or later a random jump this way or that—a jump taken not because the bowl is hot (it no longer is), but merely because the frog gets restless—will propel the frog beyond the bump, after which it will slide down to the bowl's lowest point, as in Figure 10.1b. A Higgs field can behave similarly. Its value throughout all of space may get stuck on its energy bowl's central bump while the temperature drops too low to drive significant thermal agitation. But quantum processes will inject random jumps into the Higgs field's value, and a large enough jump will propel it off the plateau, allowing its energy and pressure to relax to zero.[10] Guth's calculations showed that, depending on the precise shape of the bowl's bump, this jump could have happened rapidly, perhaps in as short a time as .00000000000000000000000000001 (10^{-35}) seconds. Subsequently, Andrei Linde, then working at the Lebedev Physical Institute in Moscow, and Paul Steinhardt, then working with his student Andreas Albrecht at the University of Pennsylvania, discovered a way for the Higgs field's relaxation to zero energy and pressure throughout all of space to happen even more efficiently and significantly more uniformly (thereby curing certain technical problems inherent to Guth's original proposal[11]). They showed that if the potential energy bowl had been smoother and more gradually sloping, as in Figure 10.2, no quantum jumps would have been necessary: the Higgs field's value would quickly roll down to the valley, much like a ball rolling down a hill. The upshot is that if a Higgs field acted like a cosmological constant, it did so only for a brief moment.

